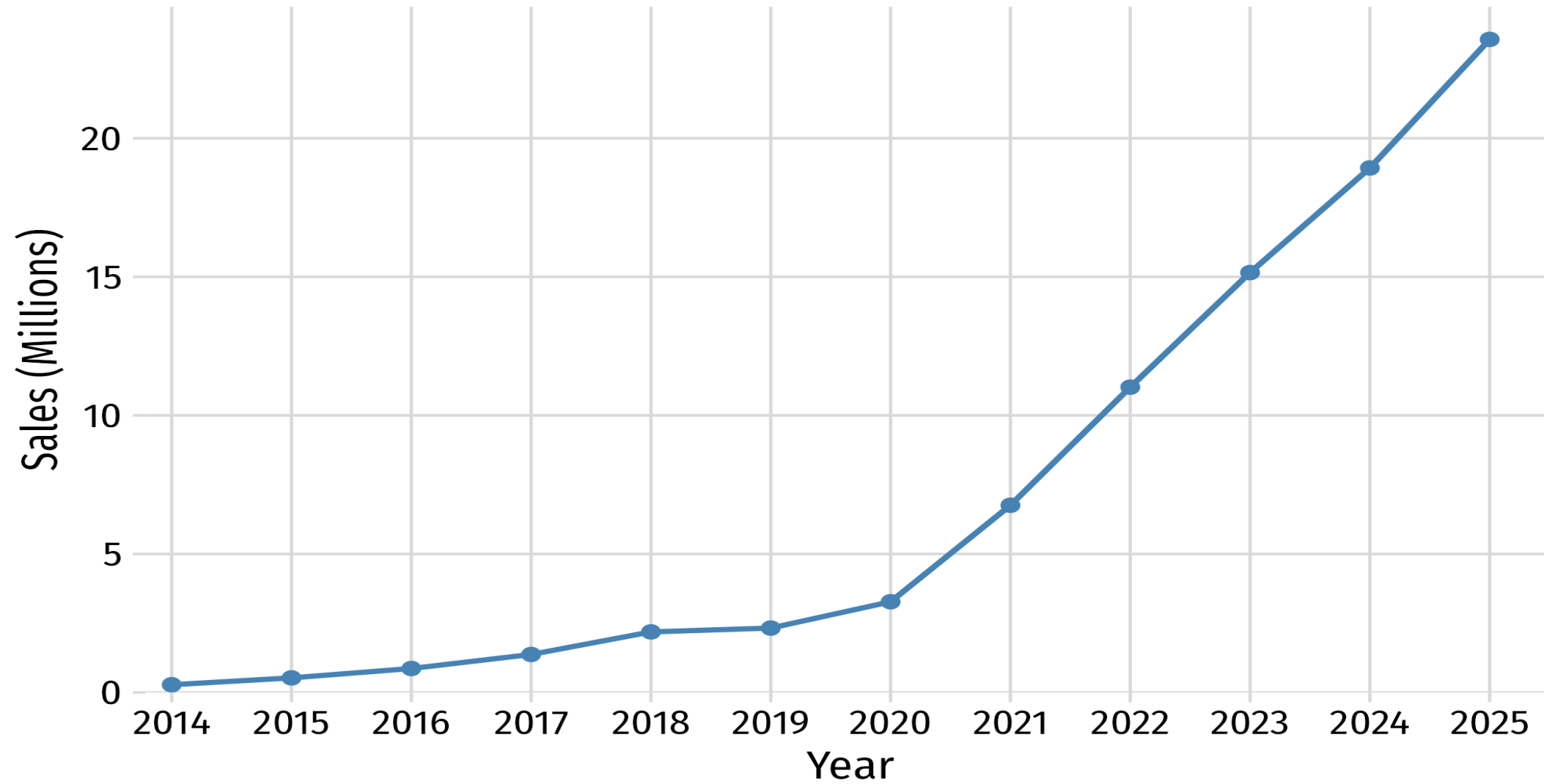


The American EV Transition: Looking Ahead While Lagging Behind (China)

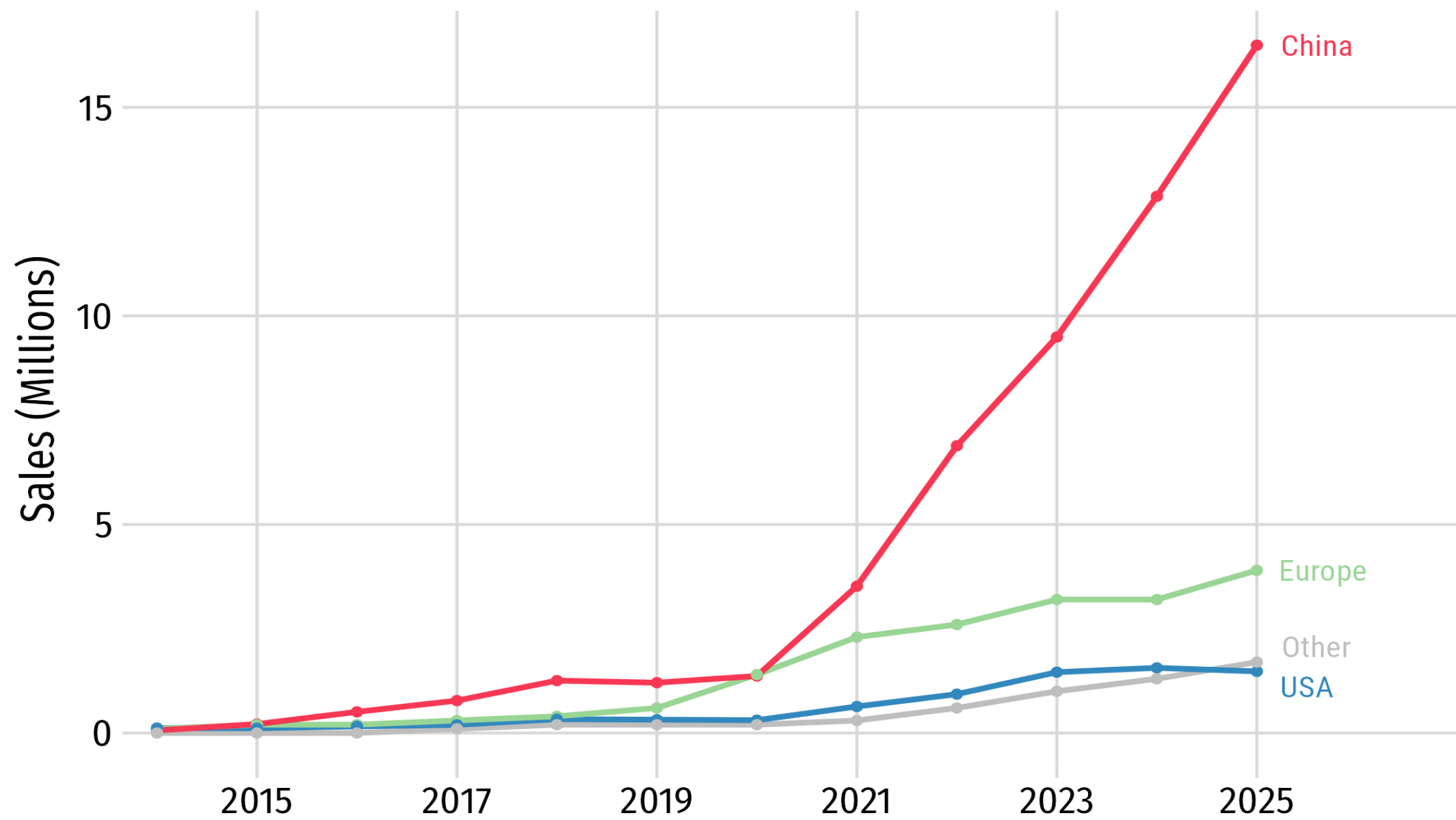


John Paul Helveston
George Washington U.
October 6, 2026

Global Annual Plug-in Electric Vehicle Sales (2014 - 2025)

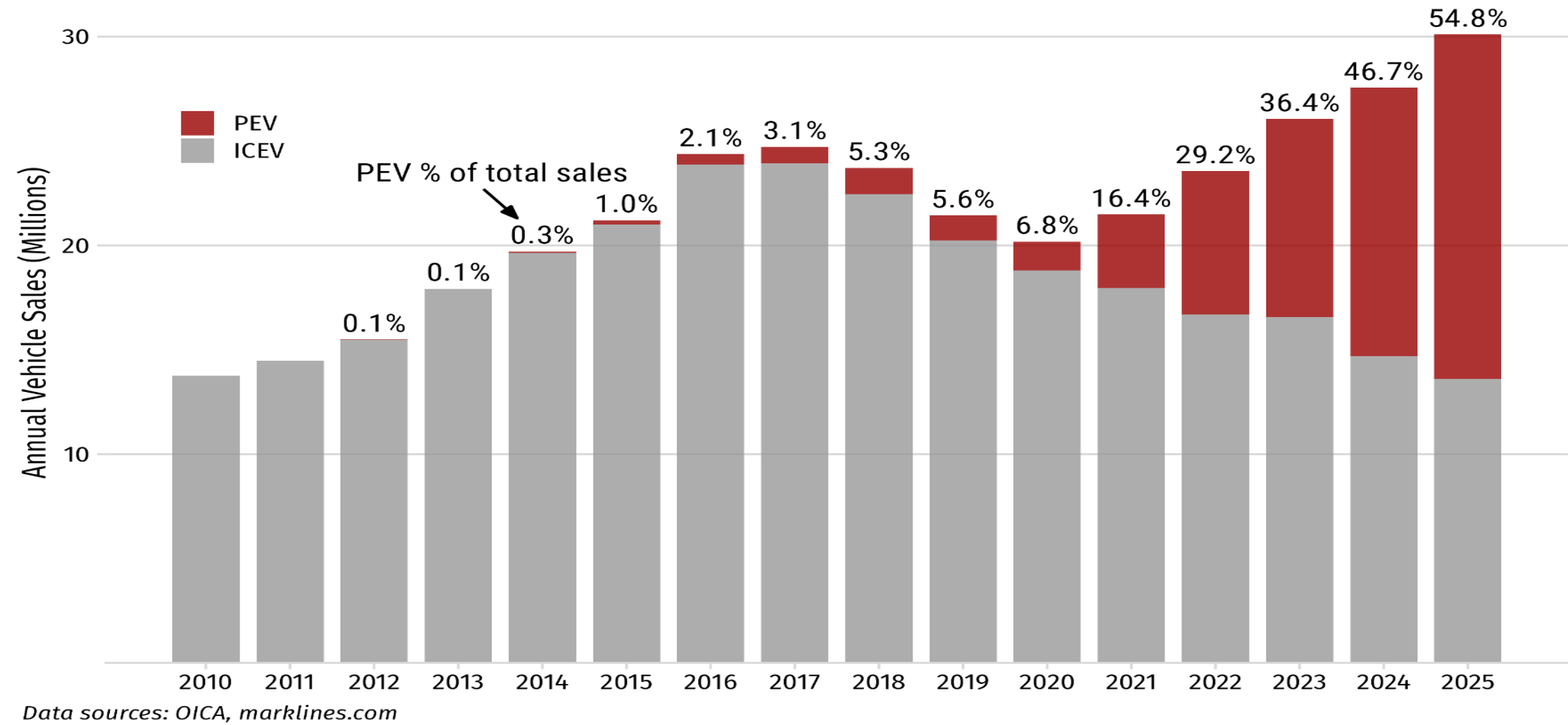


China is driving the EV revolution

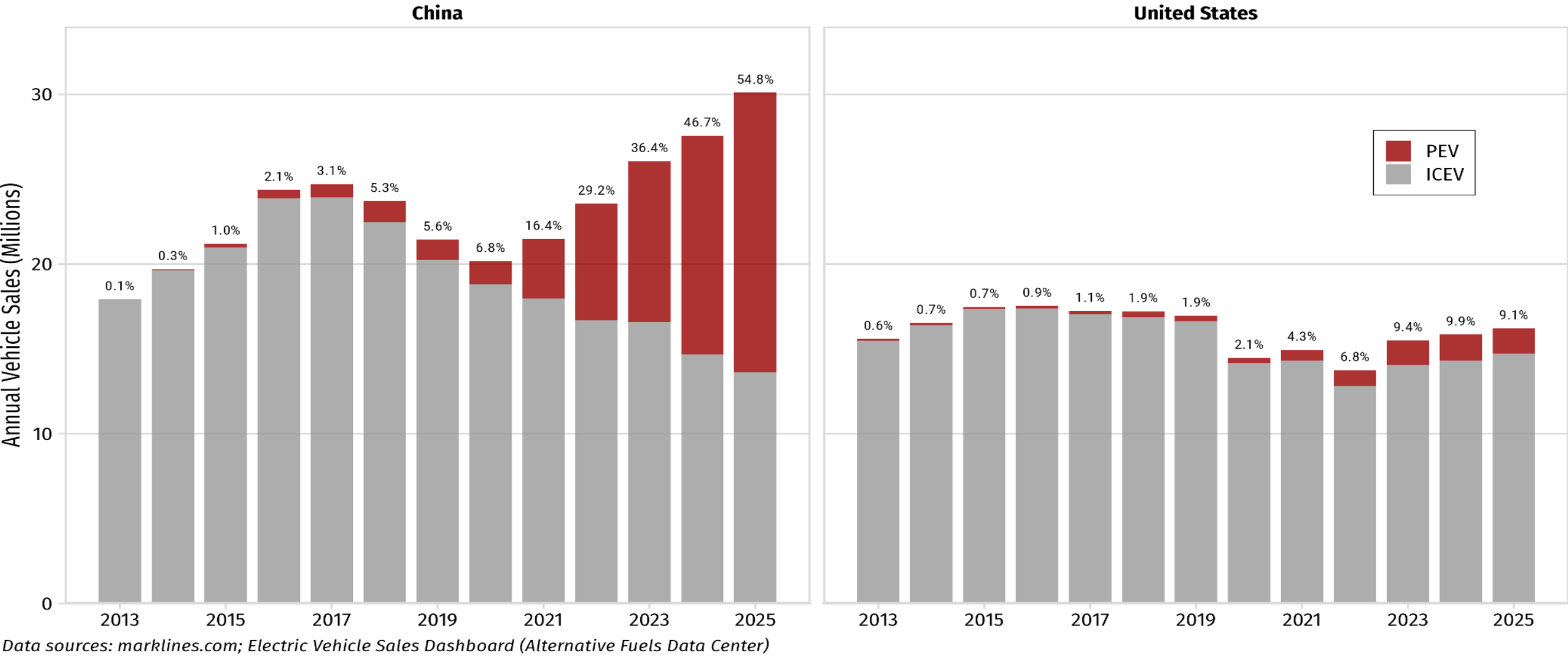


In China, PEV sales grow while ICEV sales slow

After peaking in 2017, internal combustion engine vehicle (ICEV) sales have declined for 8 straight years

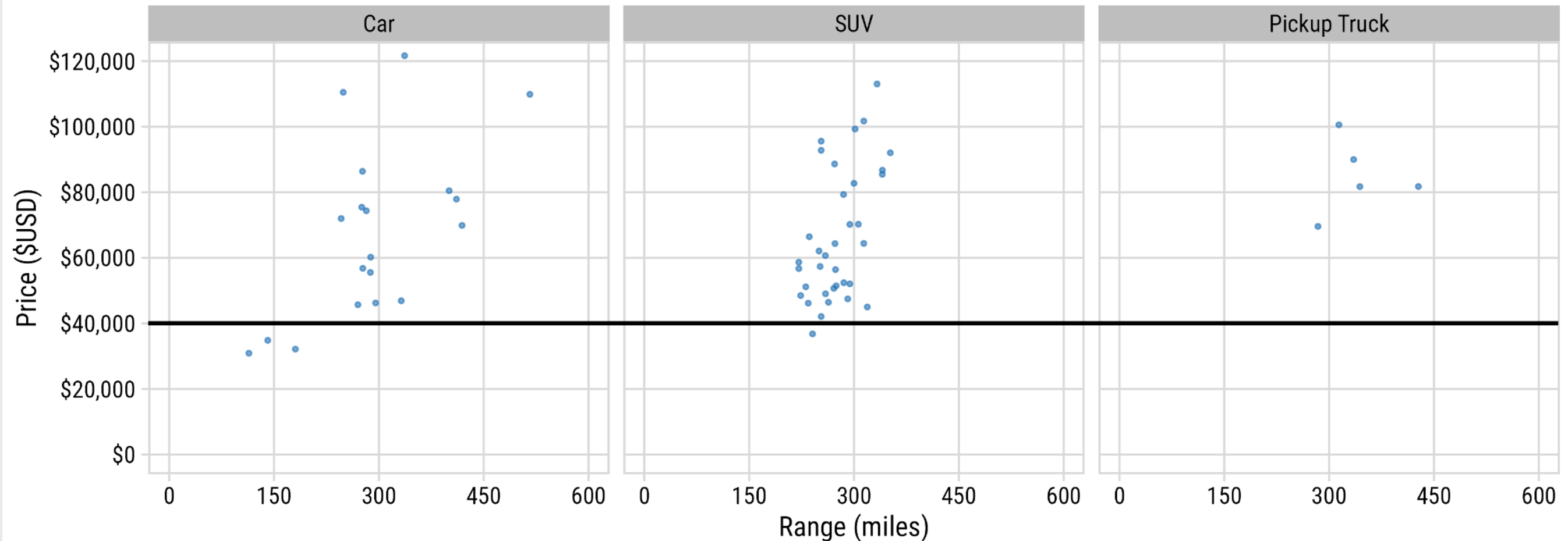


PEV adoption in China and the United States



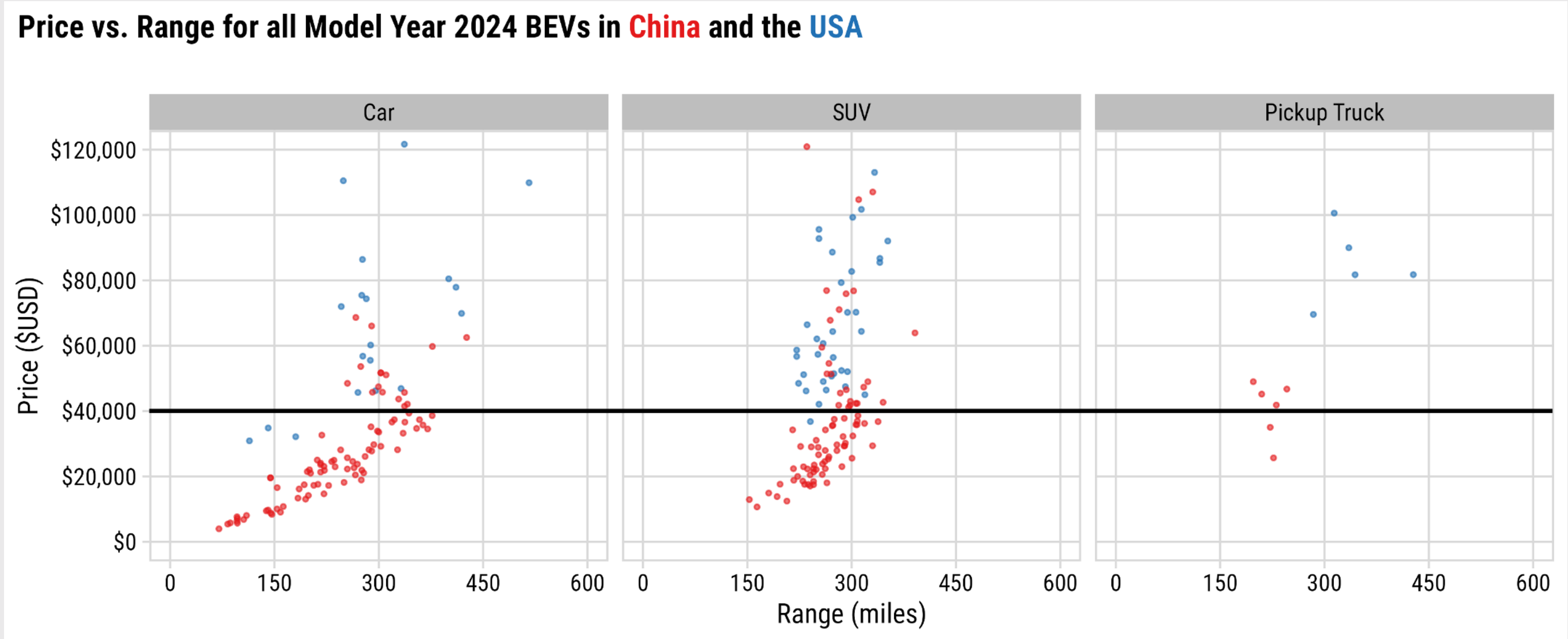
China offers more affordable BEVs across all range categories

Price vs. Range for all Model Year 2024 BEVs in **China** and the **USA**



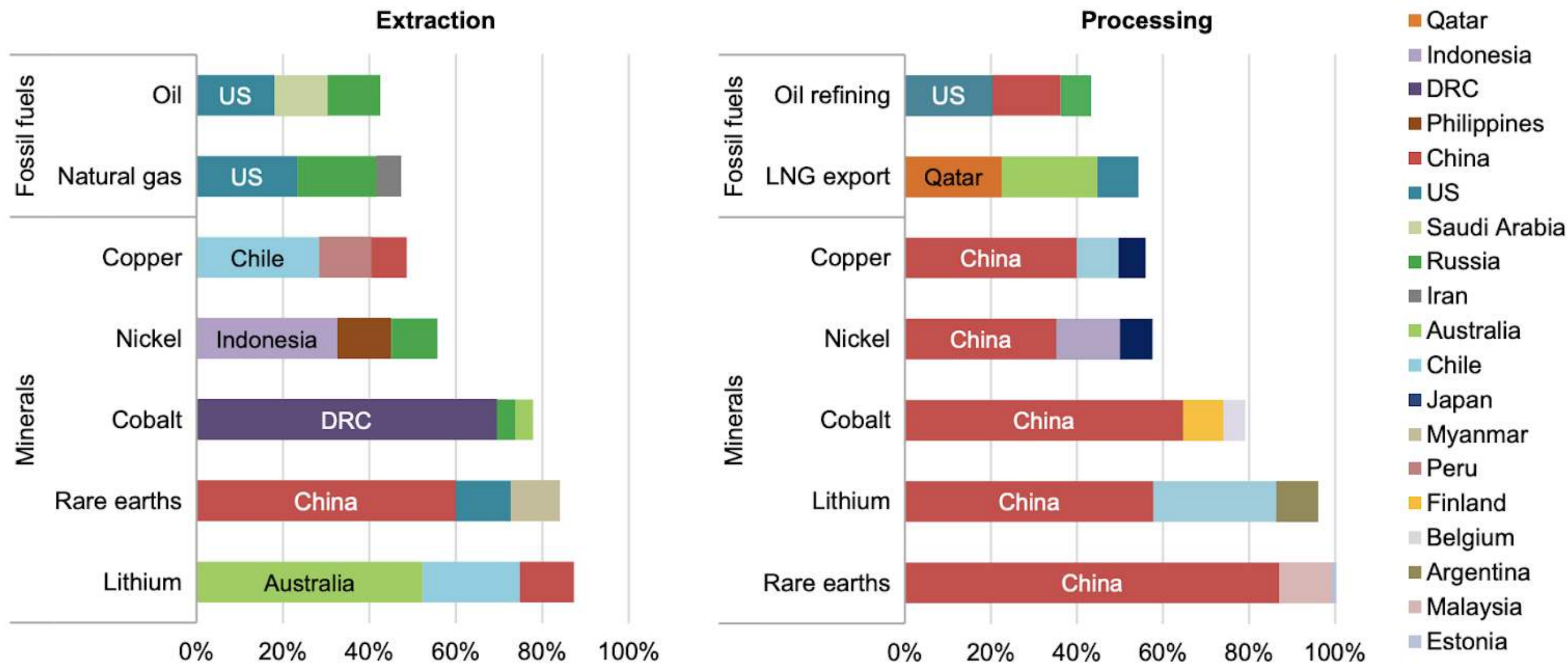
Data scraped from autocango.com (China) and carsheet.io (USA)

China offers more affordable BEVs across all range categories



Data scraped from autocango.com (China) and carsheet.io (USA)

China dominates mineral processing supply chain



How did this happen?

It's the subsidies, right?!

Estimated \$230B in subsidies (2009 - 2023)

Type of Support	2009-2017	2018	2019	2020	2021	2022	2023	Total
Rebate	37.8	4.3	3.3	3.5	7.4	9.2	0.0	65.7
Sales Tax Exemption	10.8	7.7	6.4	6.6	16.4	30.3	39.6	117.7
Infrastructure Subsidies	2.3	0.2	0.2	0.3	0.3	0.6	0.6	4.5
Research & Development	2.0	3.6	3.4	3.5	4.3	3.9	4.3	25.0
Government Procurement	7.8	1.6	1.4	2.9	1.7	1.8	0.8	18.0
Total	60.7	17.4	14.8	16.8	30.1	45.8	45.3	230.9
Spending as Share of Total Sales	42.4%	22.7%	23.3%	25.4%	18.3%	15.1%	11.4%	18.8%
Subsidy per Vehicle (US\$)	–	13,860	12,311	12,294	8,538	6,656	4,764	–

<https://www.csis.org/blogs/trustee-china-hand/chinese-ev-dilemma-subsidized-yet-striking>

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50% sales tax exemptions
30% direct rebates

<https://www.csis.org/blogs/trustee-china-hand/chinese-ev-dilemma-subsidized-yet-striking>

Institutions Matter

诸侯 经济

**Local
Protectionism**





R&D focus on production innovation

Why are *Chinese* firms making the best EVs?
(and not multinationals)

The Chinese Joint Venture System

1980s: 以市场换技术 = “Exchange market for technology”

“这就像吸食鸦片一样，一旦你沾染上了就永远也无法戒掉。”

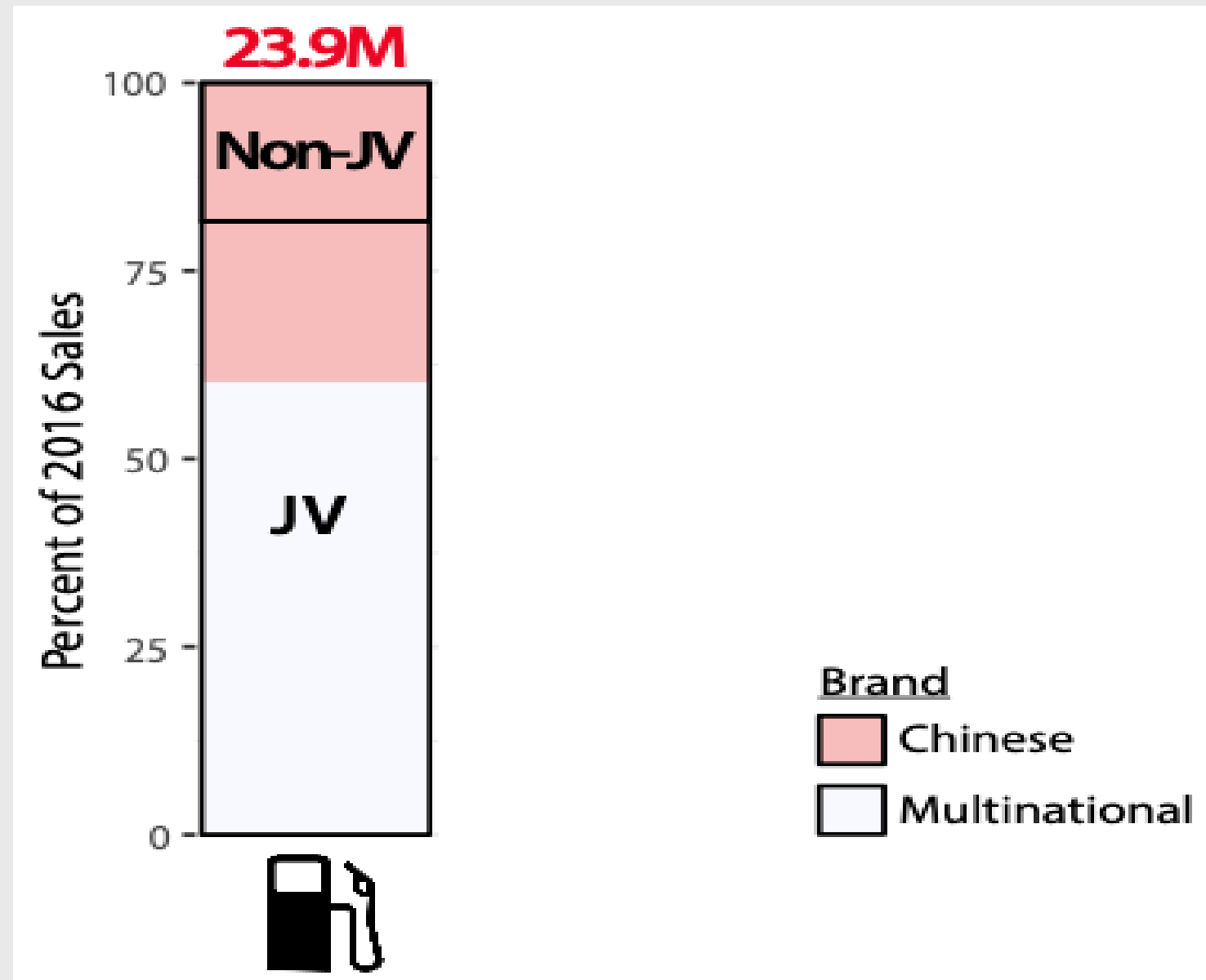
何光远, 中国前机械工业部部长

“It’s like opium. Once you’ve had it you will be addicted forever.”

Guangyuan He, Former Minister of Machinery and Industry (Reuters, 2012)

Who's going to make the EVs?

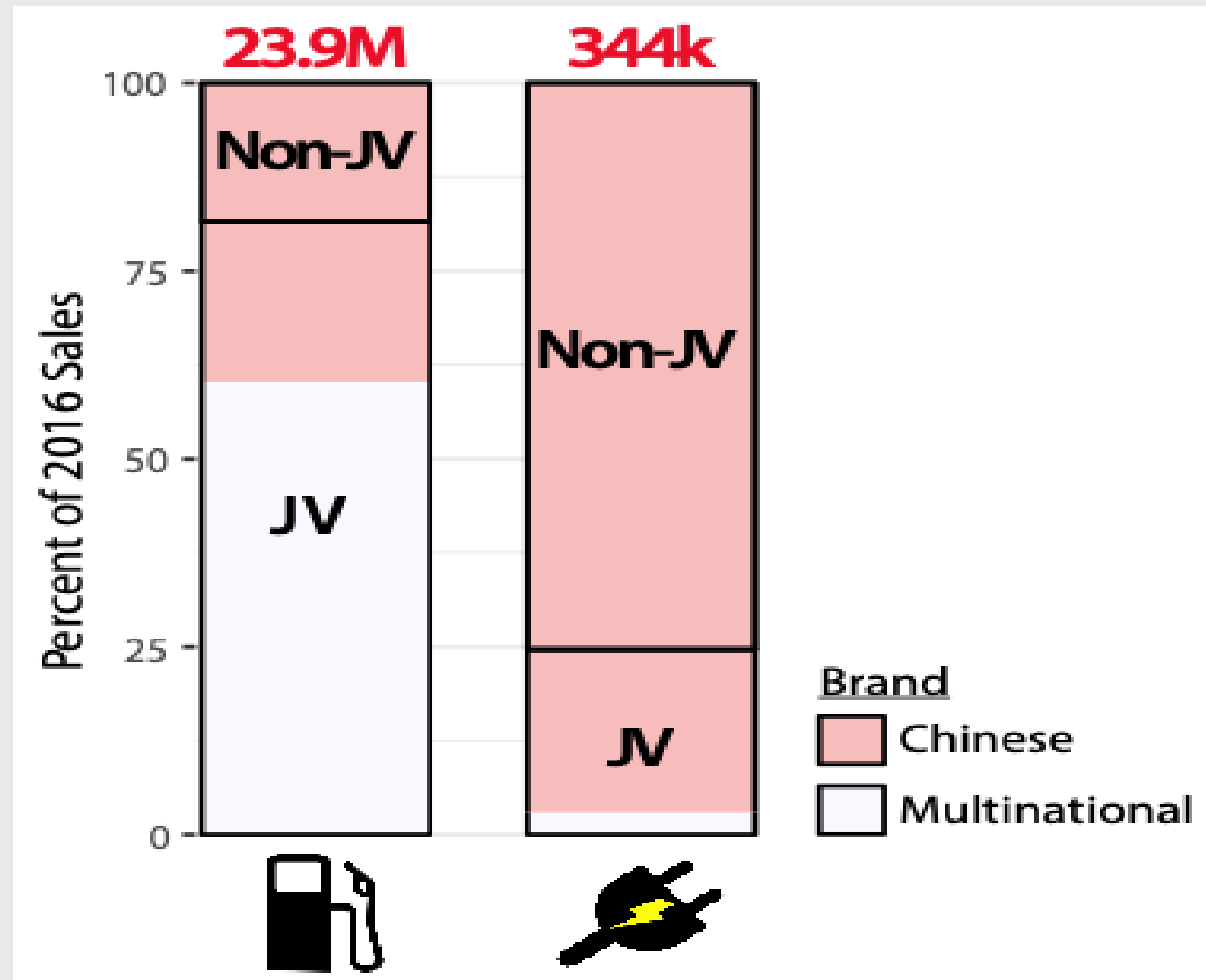
- ✗ Multinationals don't want to share cutting-edge technology
- ✗ Chinese JV partners lack incentives to innovate
- ✓ Indigenous non-JV firms



Multinational OEMs dominate traditional auto sector through large JVs

Indigenous non-JV firms dominate emerging EV sector

Helveston, J. P., Y. Wang, V. Karplus, E. Fuchs
(2018) "Institutional Complementarities: The Origins of Experimentation in China's Plug-in Electric Vehicle Industry." *Research Policy*. 48(1), pg. 206-222



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EV Challenges in the US

- 1) Policy environment
- 2) Consumer preferences
- 3) Affordability crisis

Decades of Policy Whiplash

Obama (2009–2016)

- \$7,500 EV tax credit
- Goal: 1M EVs by 2015
- DOE battery R&D investment

Trump I (2017–2020)

- Proposed eliminating EV tax credits
- Rolled back fuel economy standards
- Withdrew from Paris Agreement

Biden (2021–2024)

- IRA: \$369B in clean energy investment
- Expanded EV credits, domestic content rules
- 100% EV federal fleet goal

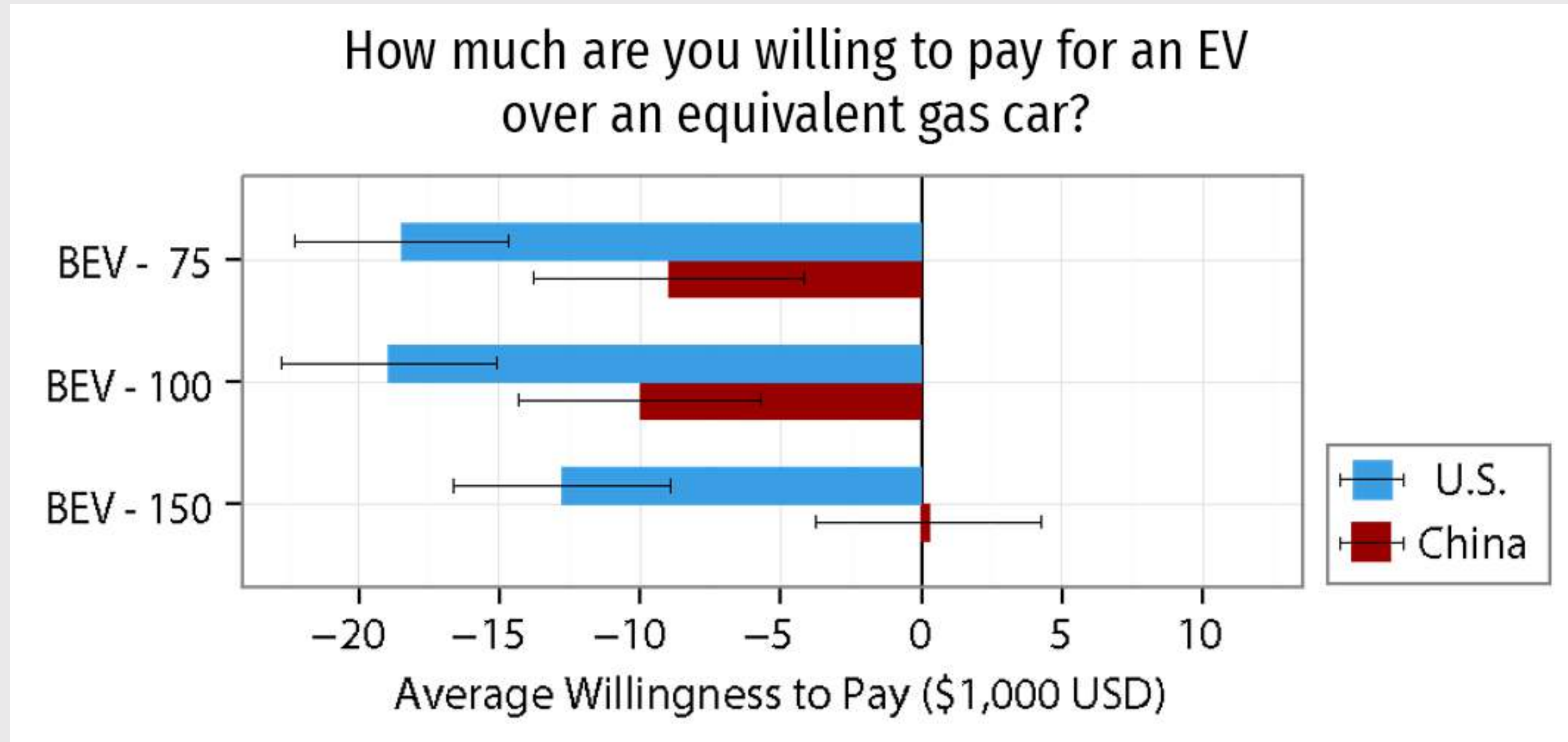
Trump II (2025–Now)

- Gutted IRA
- Rolled back fuel economy standards

EV Challenges in the US

- 1) Policy environment
- 2) Consumer preferences
- 3) Affordability crisis

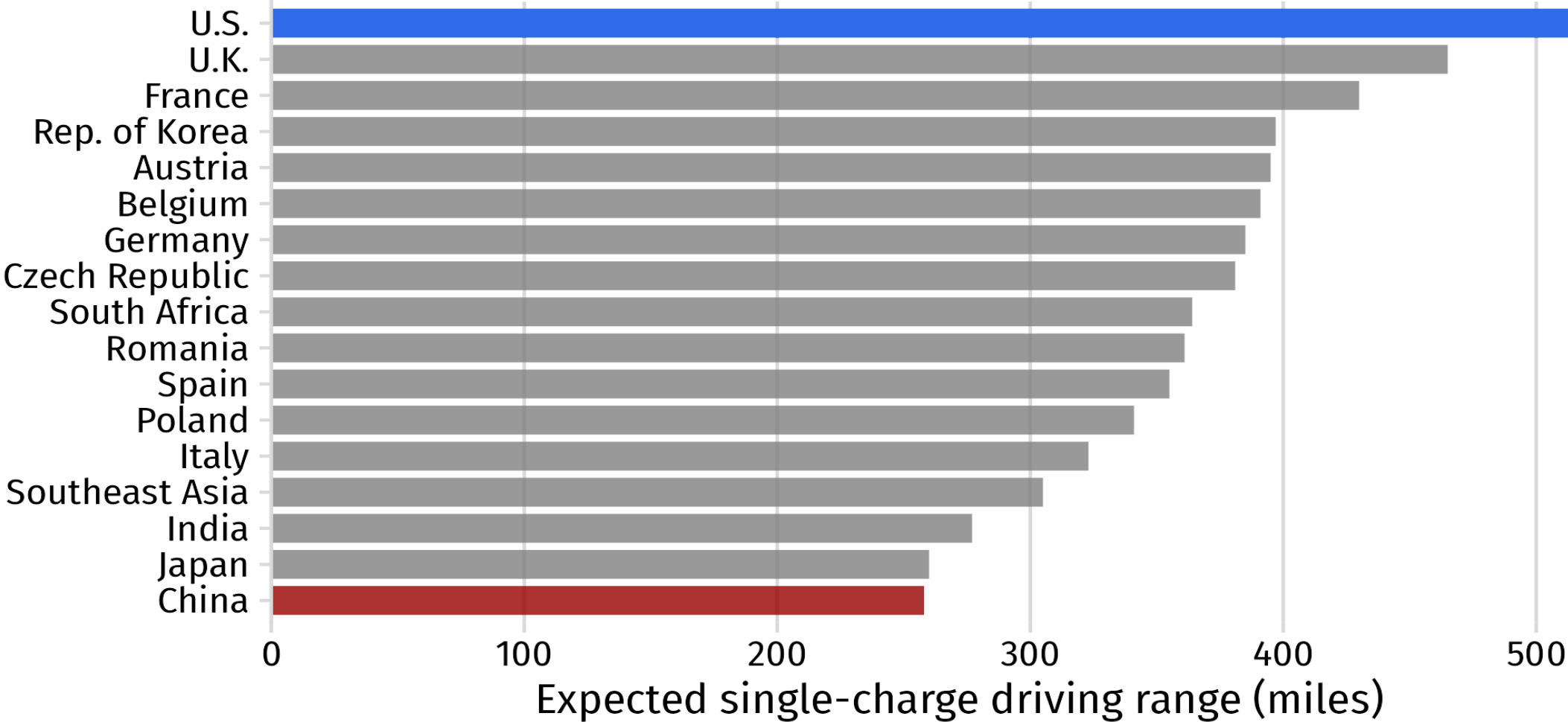
US buyers are more opposed to EVs than Chinese buyers



Helveston et al. (2015) "Will subsidies drive electric vehicle adoption? Measuring consumer preferences in the U.S. and China" *Transportation Research Part A: Policy and Practice*. 73, 96–112. DOI: [10.1016/j.tra.2015.01.002](https://doi.org/10.1016/j.tra.2015.01.002)

US buyers demand far more driving range than Chinese buyers

Average response to question: "How much driving range would a fully charged all-battery electric vehicle need to have in order for you to consider acquiring one?"



Data source: Deloitte 2022 Global Automotive Consumer Study

Top 4 Selling Vehicles in **China**

Xiaomi SU7

\$30,171 - \$41,909



BYD Song

\$24,721 - \$38,555



Geely Xingyuan

\$ 9,615 - \$13,667



Tesla Model Y

\$36,822 - \$43,809



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Tesla Model Y

\$36,822 - \$43,809



Top 4 Selling Vehicles in **USA**

Toyota RAV4

\$29,250 - \$38,955



Honda CRV

\$30,100 - \$41,100



Ford F-150

\$44,095 - \$79,005



Chevrolet Silverado

\$42,700 - \$70,000



EV Challenges in the US

- 1) Policy environment
- 2) Consumer preferences
- 3) Affordability crisis



OPINION
GUEST ESSAY

Where Did All the Affordable Cars Go?

 Share full article



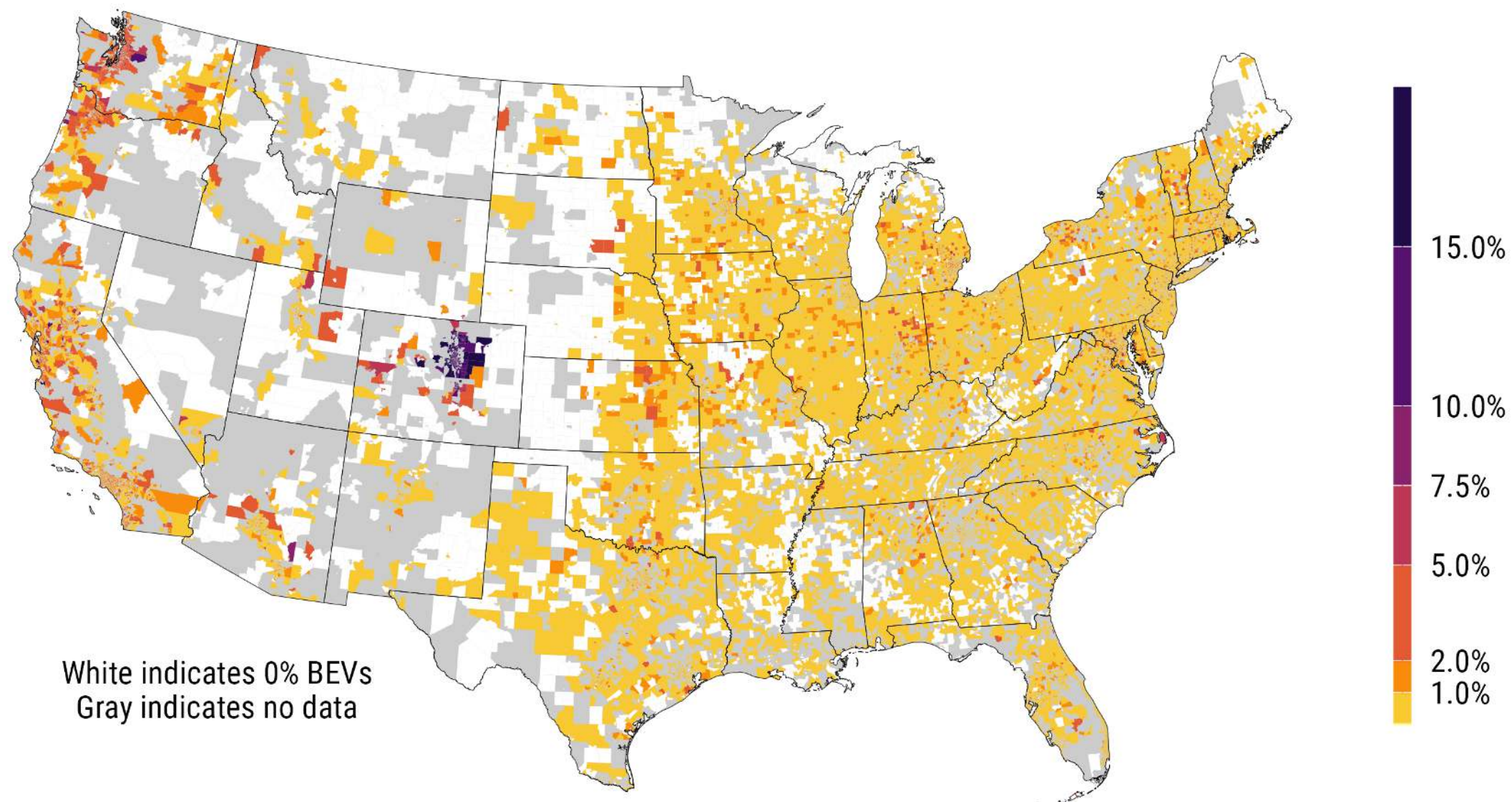
  3.2K

By Clifford Winston

Mr. Winston is an economist who specializes in transportation and microeconomic policy.

April 13, 2026

The EV “Deserts” of America

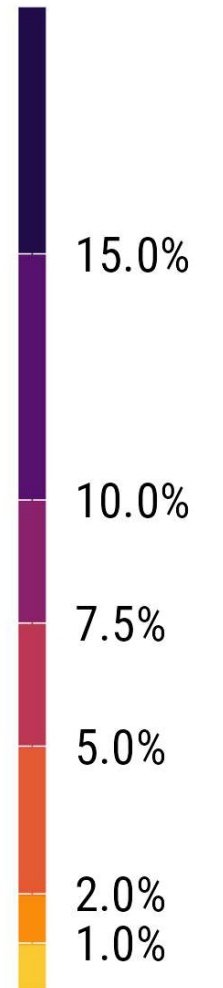
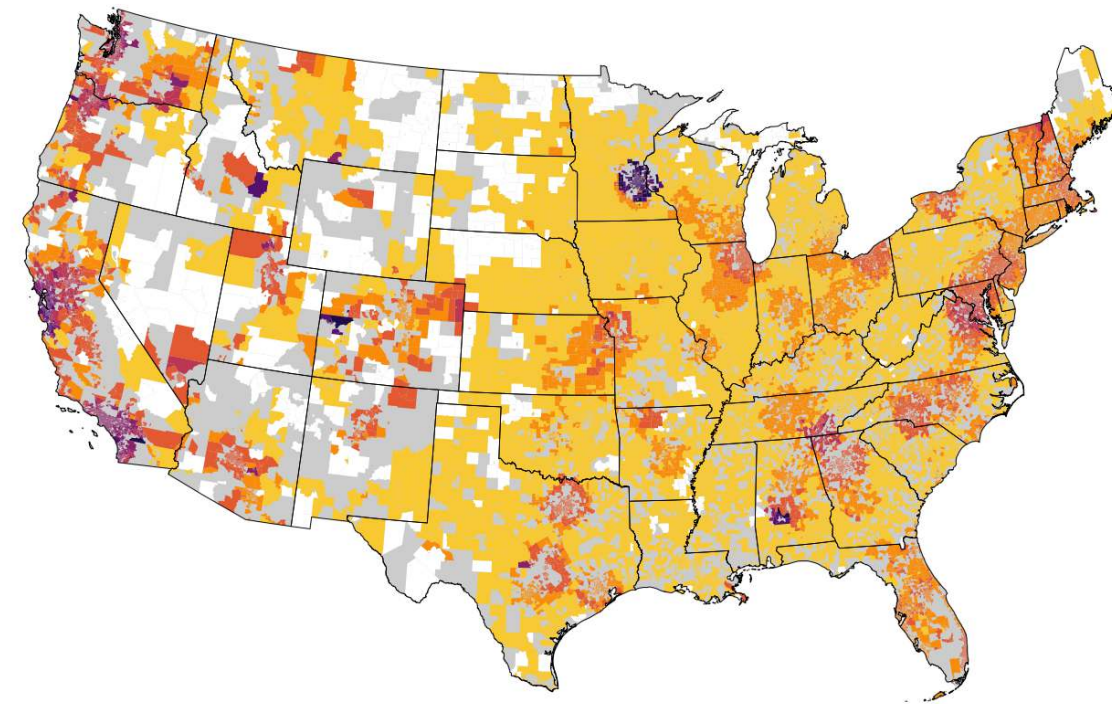
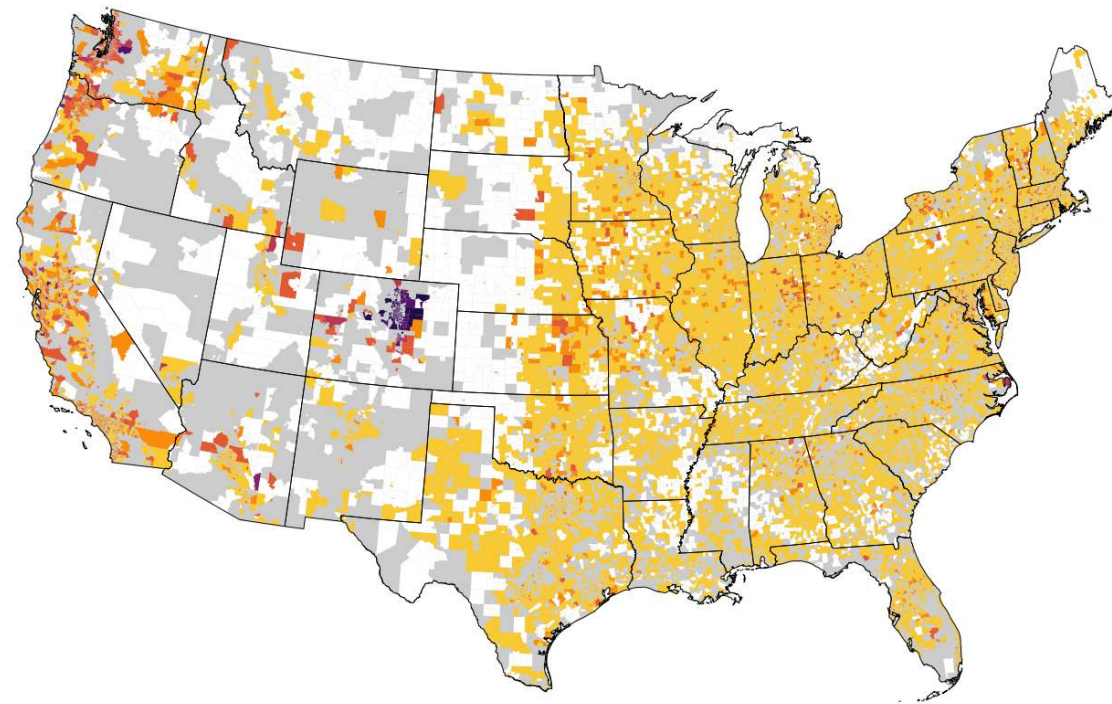


The EV “Deserts” of America

White indicates 0% BEVs
Gray indicates no data

New

Used



Data: **128M vehicle listings** from 60k dealerships (2016 - 2025), marketcheck.com

How should the US respond to Chinese EVs?

Bipartisan Response: **Ban Chinese tech**

Lutnick Shuts Down Talk of Any Chinese Investment in US Autos



Howard Lutnick at the Semafor World Economy Summit in Washington on April 17. Photographer: Aaron Schwartz/Bloomberg

By Colum Murphy

April 17, 2026 at 5:50 PM EDT

Block imports: Steep tariffs on imported Chinese EVs, batteries (First Biden, continued w/Trump)

Block investment: Stricter Foreign Entities of Concern (FEOC) rules

Ban the tech: Connected Vehicle Security Act of 2026 bans any connected vehicles with Chinese tech (import or domestic)

Option A

Exclude Chinese EVs

- China “cheated” its way to the top with subsidies
- Chinese EVs are a security threat

Option B

Engage strategically

- China built real competitive advantages in manufacturing and supply chains
- Chinese EVs are winning markets, US OEMs are losing markets

The Tesla Example: Transparent Cybersecurity

China allowed Tesla in via
targeted compliance:

- Store vehicle data **within China**
- Restrict operations near **sensitive areas**
- Use only **government-approved** mapping systems (Baidu)



The European Model: Block Imports, Welcome Chinese FDI

Access to cutting-edge battery tech

Create local jobs

Increase supply chain resilience

CATL

\$7.6B battery plant in Debrecen
100 GWh/year for Mercedes-Benz, BMW,
Volkswagen

BYD

First European EV manufacturing facility
Producing BEVs and PHEVs for the
European market

US Opportunities

Technology Licensing Agreements

Ford-CATL: Licensing battery technology in a Michigan plant

Challenge: CATL recently added to DOD's list of "Chinese military companies"

Chinese FDI into U.S.

Gotion batteries: Multi-billion dollar investments in Illinois and Michigan

Challenge: Uncertainty around Foreign Entities of Concern (FEOC) status

Consumers aren't waiting for Washington

Public Opinion



Topics ▾Regions ▾Events ▾Publications ▾Become a

Home > Topics/Regions >

Almost 70% of people globally are comfortable with their government buying Chinese-made clean technologies, new survey finds

Exclusive: Trump pollster finds MAGA voters like solar energy



Ben Geman



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Electric Vehicles: Gas Price Tipping PointHybrids' New PopularityCutting Oil DemandE-Bikes Hit a BumpChina's EV Battery

IndustriesTransportation

Canadians Say They'll Buy Cheaper Chinese EVs as Tariffs Drop

COX AUTOMOTIVE

Brands ▾What We Offer ▾Insigh

1. Market polarization defines opportunity — and risk

Consumer sentiment toward Chinese auto brands is sharply divided. Younger, EV-oriented shoppers show meaningful openness, while older and domestic-loyal buyers remain resistant.

Consideration of Chinese Brands in the U.S.

38%
extremely/very likely

39%
not very/not at all likely

69%
of Gen Z more likely

Ford's CEO Doesn't Want To Stop Driving China's Apple Car

Ford CEO Jim Farley has been driving an imported Xiaomi SU7 for months. He says it's fantastic.

"We flew one from Shanghai to Chicago, and I've been driving it for six months and I don't want to give it up," - Ford CEO Jim Farley



The U.S. risks becoming an **“island of tailpipes”**

Strategic Responses

 **North American Integration:** Build sufficient scale through Canada and Mexico partnerships

 **Transparent Security Standards:** Develop compliance frameworks all connected vehicles must meet — *regardless of origin*

 **Battery Manufacturing Focus:** Prioritize domestic production as platform for commercializing next-gen U.S. technologies

 **Strategic Chinese Partnerships:** Pursue licensing agreements and FDI with targeted safeguards — not wholesale restrictions

 **Sustained Policy Support:** Restore and maintain IRA incentives long enough for investments to mature

Thanks!

Slides available at jhelvy.com/slides

jhelvy.com 

jph@gwu.edu 

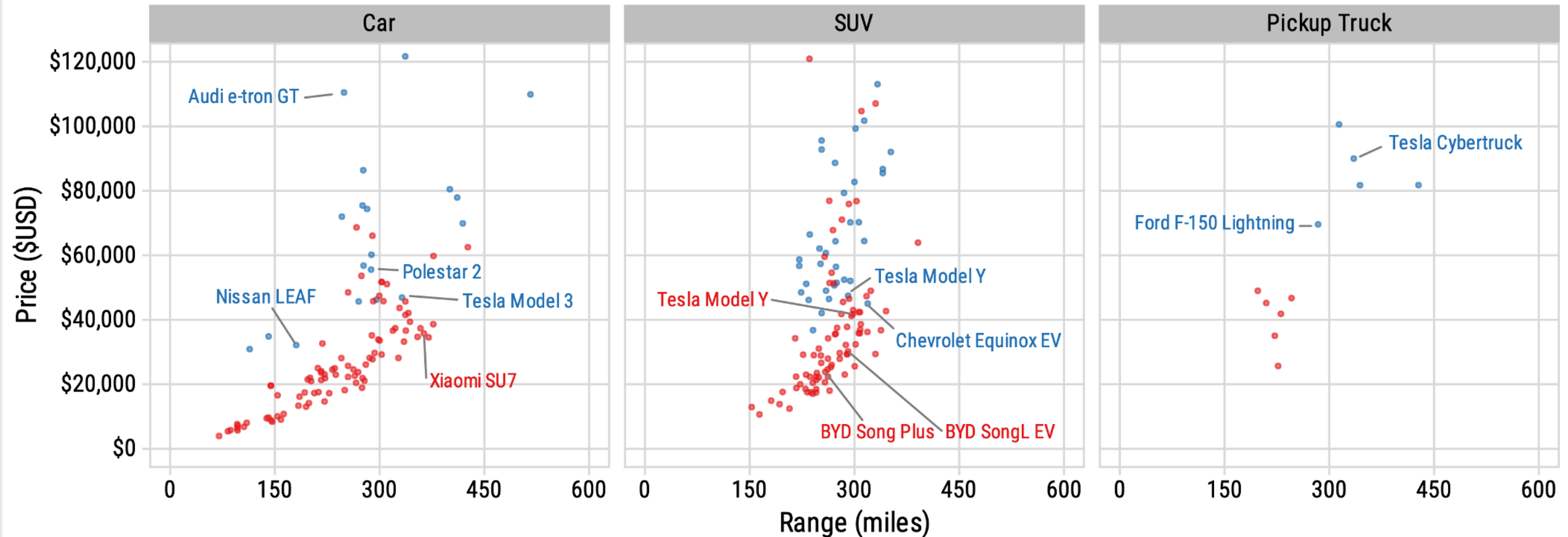
[@jhelvy](#) 

Extra Slides

China's EV subsidies are on par with typical subsidies
needed to start an industry

China offers more affordable BEVs across all range categories

Price vs. Range for all Model Year 2024 BEVs in **China** and the **USA**

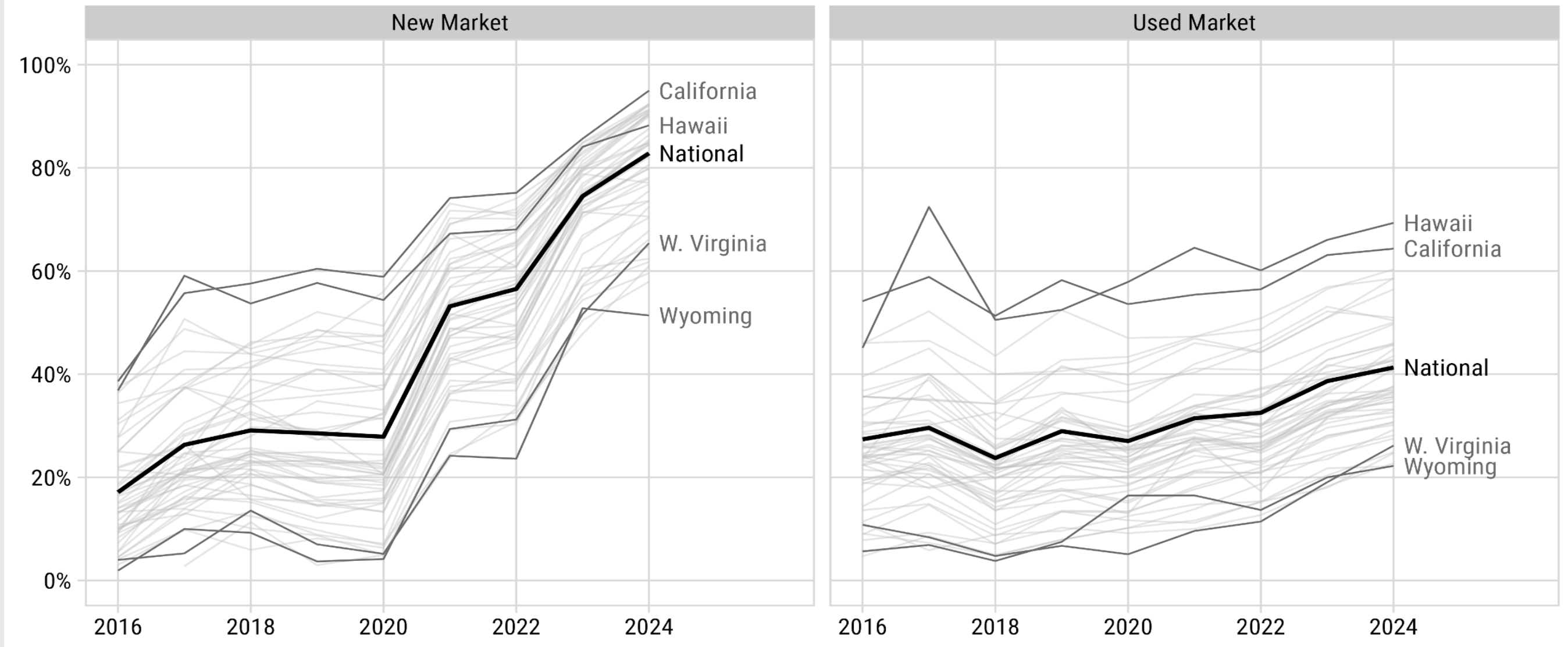


Data scraped from autocango.com (China) and carsheet.io (USA)

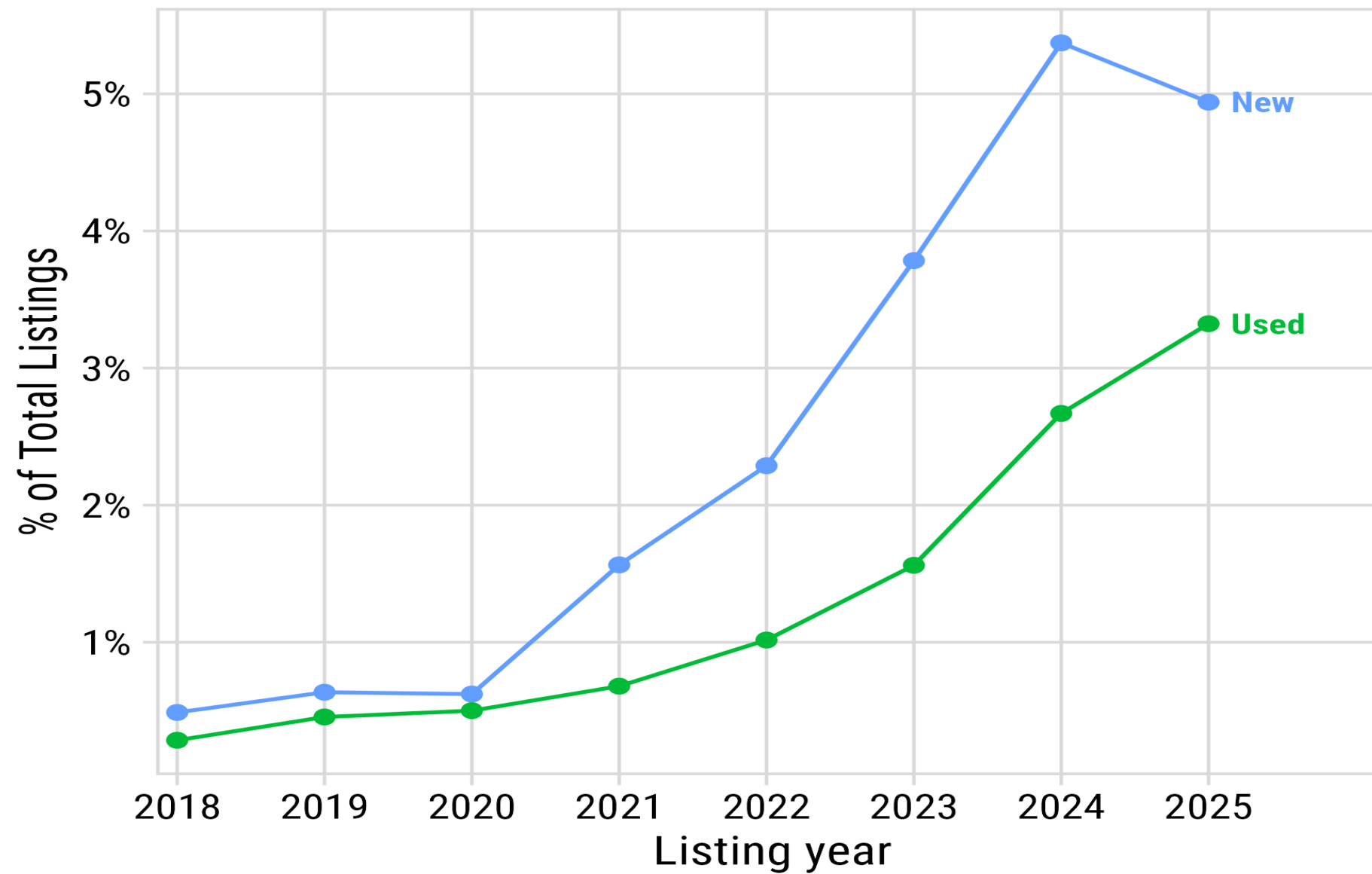
How many dealerships are carrying EVs?

4/5 dealers have *new* BEV; 2/5 dealers have *used* BEV

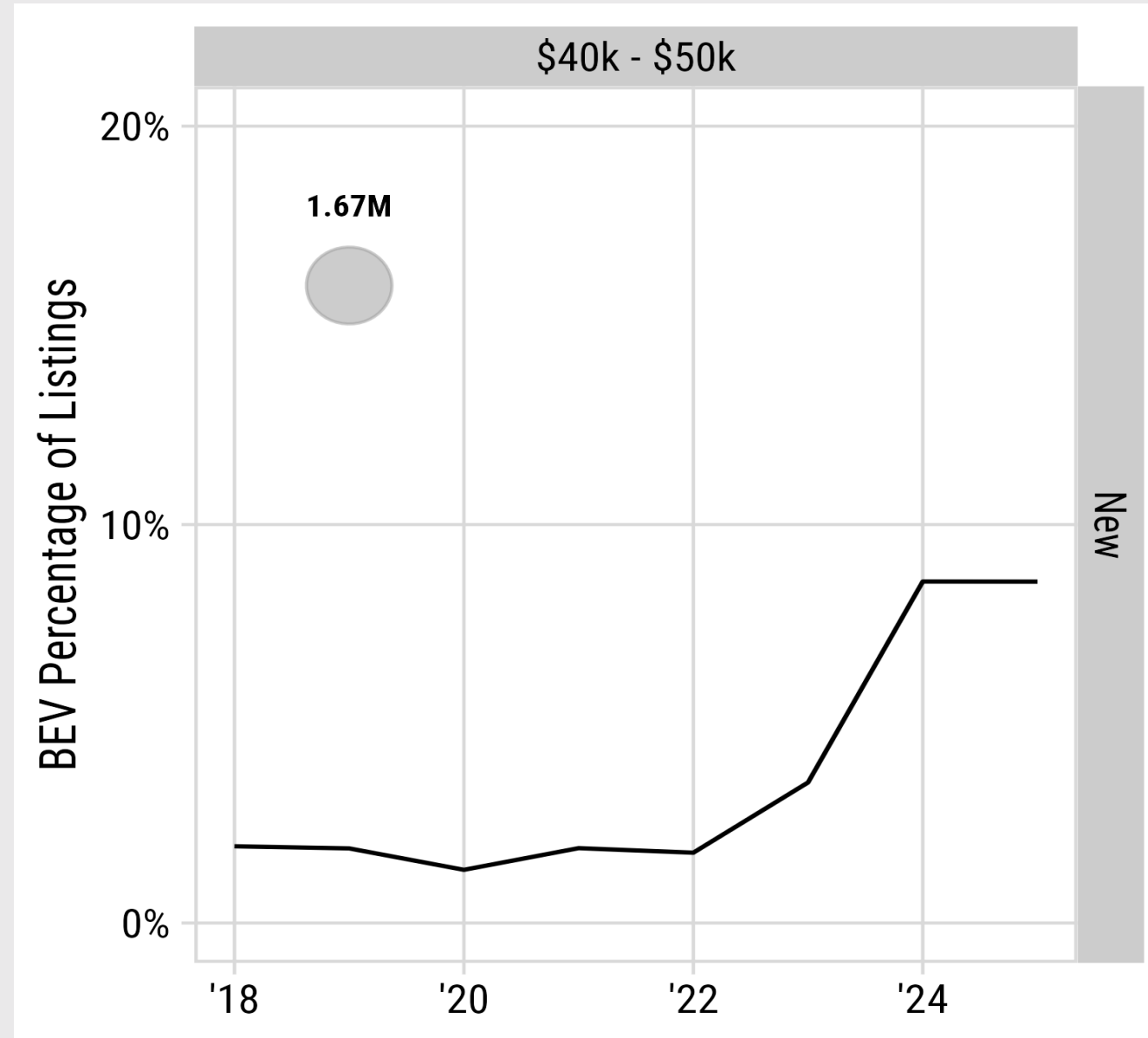
Percentage of dealerships with at least one PEV listing



Most dealers
carry very few
EVs



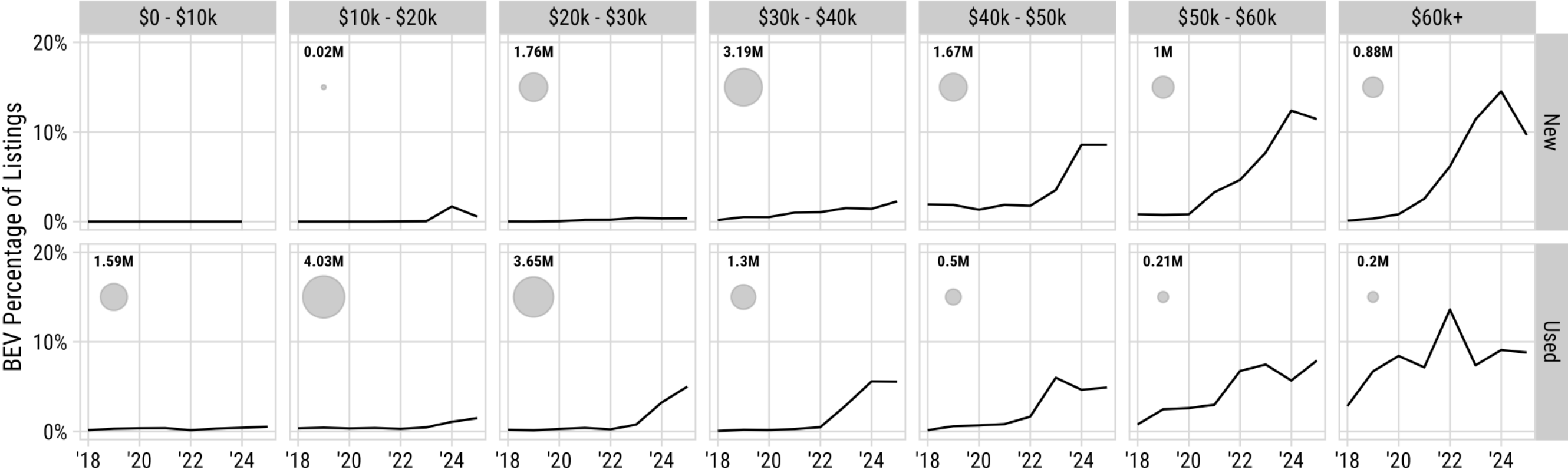
EV percentage of vehicle listings for \$40-50k price bin



Majority of BEV listings in high-price segments

BEV percentage of vehicle listings in new and used markets by price bin

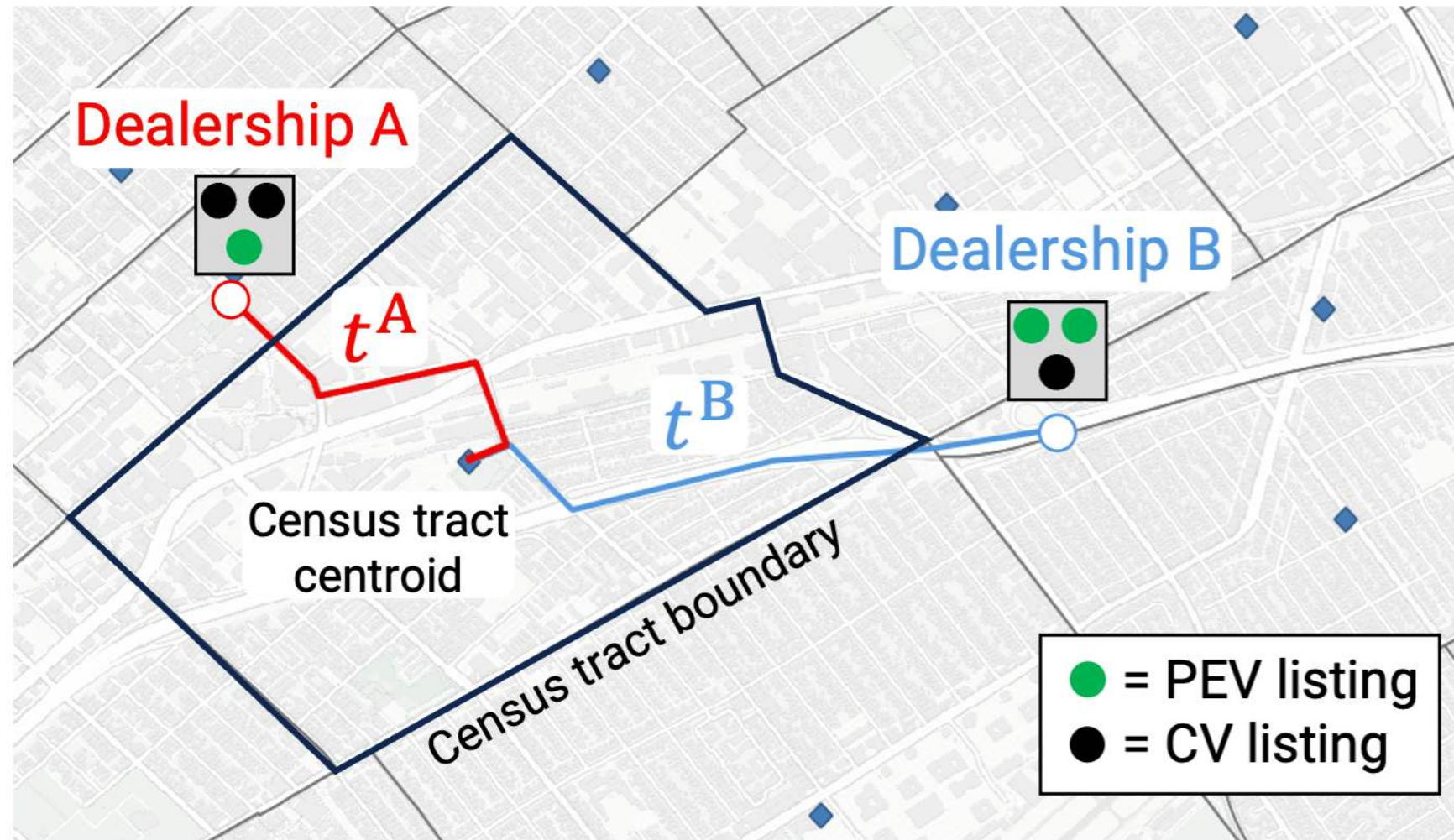
Bubbles show total 2025 listings (in millions) for each market and price bin



How hard is it to get to a PEV dealer?

Vehicle accessibility metric:

Road travel time from census tract centroid to nearest dealership with a target vehicle



Times to first 3 ● (PEVs)

$$t_1^{\text{PEV}} = t^A$$

$$t_2^{\text{PEV}} = t^B$$

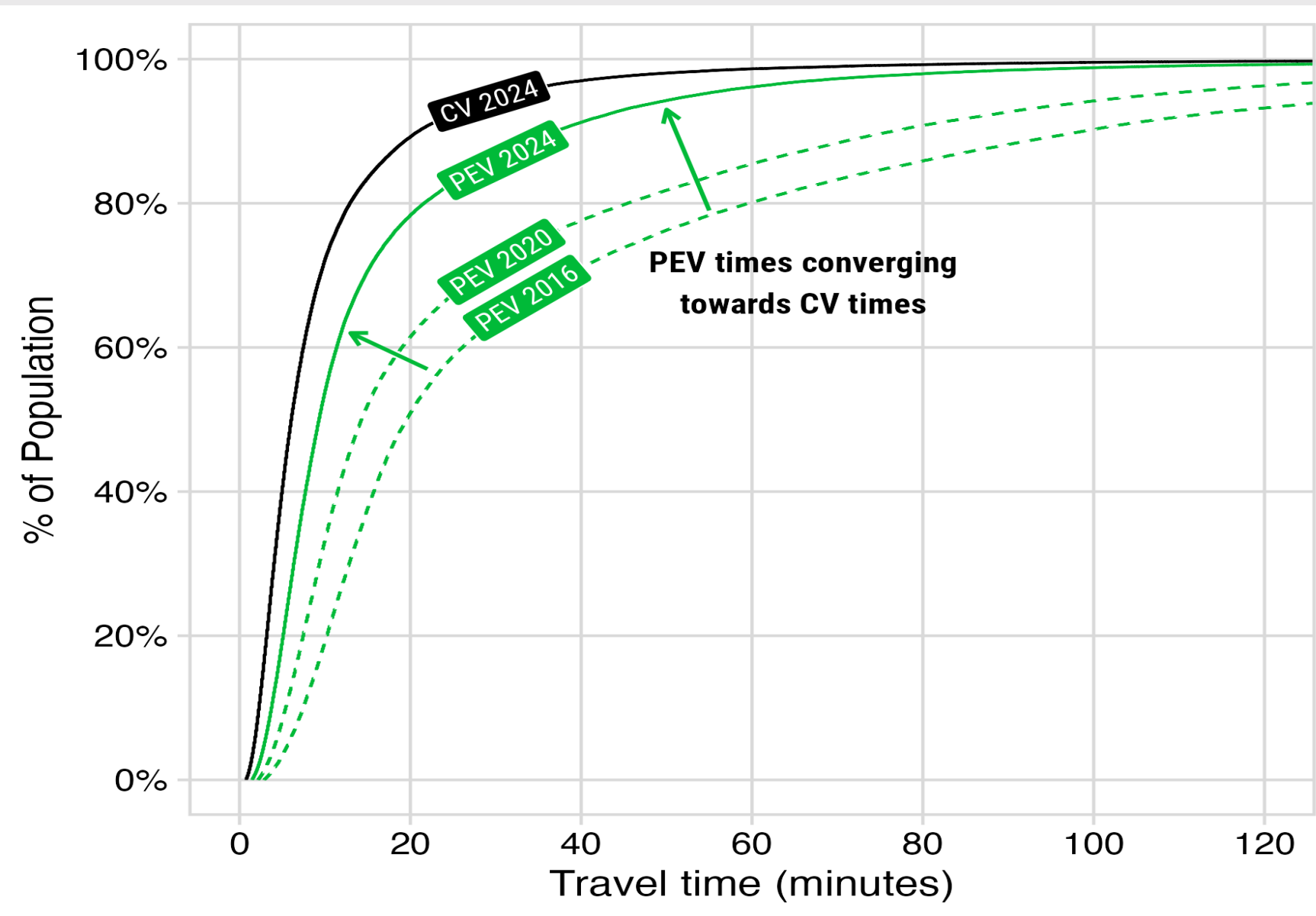
$$t_3^{\text{PEV}} = t^B$$

Times to first 3 ● (CVs)

$$t_1^{\text{CV}} = t^A$$

$$t_2^{\text{CV}} = t^A$$

$$t_3^{\text{CV}} = t^B$$



PEV travel times
are converging
towards
conventional
vehicle times

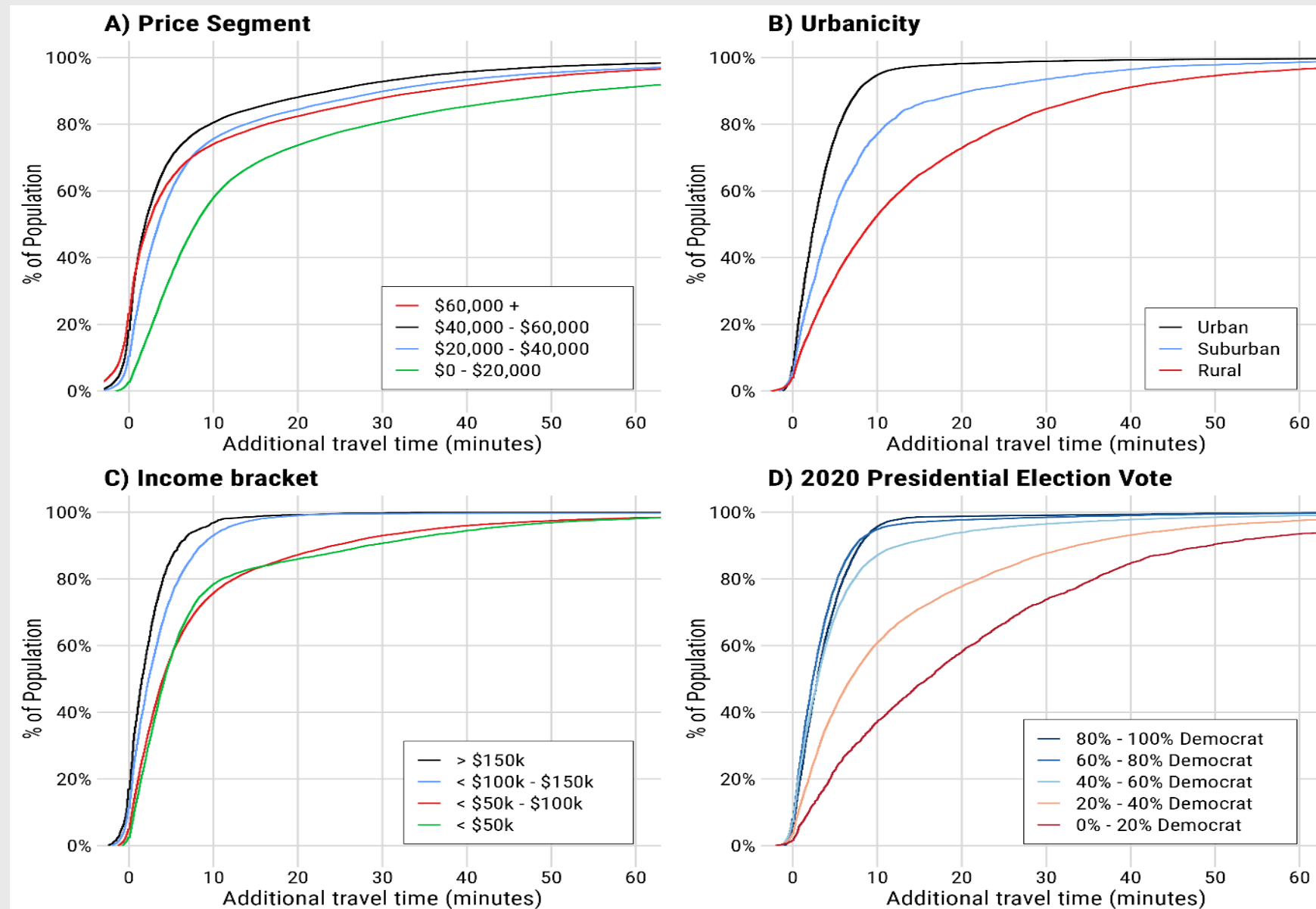
80% of pop:

- CV in ~12 min
- PEV in ~22 min (2024)
- PEV in ~60 min (2016)

Additional travel times (PEV - CV) by demographic blocks shows disparities

Places with worse PEV access:

- Lower income areas
- Rural areas
- Republican strongholds



Local vehicle market: Vehicles in 90 minute driving isochrone

Combine ML algorithm of predicted household vehicle budget (using Census data) with local market characteristics

